STAT 2102 · Practice Midterm

Unofficial practice test in the style of Homework 2 · 50 points · suggested time 75 minutes

Instructions. Answer every question in one Quarto (.qmd) file and render it to PDF. Show the code and its output for each part, and answer the question **in words** below the output. Use `set.seed()` where a question tells you to. Use functions from the apply family (or vectorized code) instead of loops where you can. Check your course's exam rules before using notes or other tools. The answer key is a separate file, so try everything first.

Problem 1: Functions (20 points)

Each answer needs only a few lines of code.

1.1 (5 pts)

Write a function named **roll_until_doubles** that takes no arguments. It should roll two fair six-sided dice repeatedly until both dice show the same number (doubles), and return how many rolls it took. Run your function once after `set.seed(1)`. Then use `replicate()` to estimate the average number of rolls over 10,000 runs. What value would you expect, and why?

1.2 (5 pts)

Write a function named **spin** with one argument, `n`, that simulates `n` spins of a spinner with these probabilities: Red 0.50, Blue 0.30, Green 0.15, Gold 0.05. The function should return a list containing (a) a frequency table of the outcomes that shows **all four colors**, even a color that never came up, and (b) the proportion of spins that landed on Red. Test it with `set.seed(2)` and `n = 12`.

1.3 (5 pts)

Write a function named **letter_grade** that takes a numeric vector of scores and returns a vector of letter grades without using a loop: "A" for 90 and above, "B" for 80 up to (but not including) 90, "C" for 70 up to 80, and "F" below 70. Test it with `c(95, 83, 70, 64, 89.5)`.

1.4 (5 pts)

Write a function named **mean_quartiles** with arguments `n`, `mu`, `sigma` and `reps` (default 1000). It should generate `reps` samples, each of `n` observations from a normal distribution with mean `mu` and standard deviation `sigma`, compute the mean of each sample, and return a **named vector** with the **first quartile, median, third quartile and interquartile range (IQR)** of the simulated sample means. Test it with `set.seed(3)`, `n = 25`, `mu = 70`, `sigma = 10`. Without running more code, how would you expect the IQR to change if `n` were 100 instead of 25?

Problem 2: The coffee shop data (30 points)

The file `coffee_shops.csv` contains information on 400 coffee shops collected by a regional coffee company that wants to understand what drives a shop's revenue. The variables are:

Variable	Type	Description
`Shop_ID`	Numeric	Unique identifier of the shop
`Revenue`	Numeric	Monthly revenue (in thousands of USD)
`Price`	Numeric	Average drink price (USD)
`Staff`	Numeric	Number of employees
`Age`	Numeric	Median age of the shop's customers (years)
`Location`	Categorical	Campus, Downtown, Suburb
`DriveThru`	Categorical	Whether the shop has a drive-thru (No, Yes)
`Chain`	Categorical	Whether the shop belongs to a chain (No, Yes)
`Rating`	Categorical	Customer rating (Poor, Average, Excellent)

No file? Run the code in the Appendix: it creates exactly the same data and saves `coffee_shops.csv` in your working directory.

2.1 (1 pts)

Import `coffee_shops.csv` into R, converting all character variables into factors, and display the structure of the data.

2.2 (3 pts)

What is the average monthly revenue of the shops? How does the average revenue of chain shops compare to that of independent shops? Additionally, among **Campus** shops only, how does the average revenue of chain shops compare to that of independent shops?

2.3 (2 pts)

Which shop charges the highest average drink price (give its index in the data set)? Conversely, which shop has the lowest revenue?

2.4 (3 pts)

What is the median revenue for each level of customer Rating? Answer the same question again, now also taking DriveThru into account, which gives six combinations.

2.5 (4 pts)

Quartiles. Find the first, second and third quartiles of Price. Compare the average revenue of shops priced **above the third quartile** with that of shops priced **below the first quartile**. Then add a new factor variable named `Price_Group` with levels Low (Price at or below Q1), Middle (above Q1, up to Q3) and High (above Q3), without modifying Price, and report how many shops fall in each group.

2.6 (3 pts)

Compare monthly revenue across the three locations using boxplots. Customize the boxplots by adding labels, changing the colors and adding notches. Are there any outliers?

2.7 (3 pts)

Provide an overview of customer ratings using a barplot. Display the bars in the order Poor, Average, Excellent, make the bars horizontal, and customize the plot with labels and colors.

2.8 (3 pts)

Create side-by-side barplots showing the frequency of each customer Rating for shops with and without a drive-thru. Include a legend, and customize the plot as in the previous question.

2.9 (4 pts)

Plot Revenue against Price. Label the plot and axes properly. Use triangular points of size 0.9, colored by Location (one color for each of the three locations), and include a legend. Describe the relationship you see.

2.10 (4 pts)

Combine a histogram of Revenue (about 10 bins) and a boxplot of Staff into a single image. Customize both plots as in Question 2.6 (no notch on the histogram).

Appendix: create the data set

Running this code once creates the same 400 shops every time (because of `set.seed`) and writes `coffee_shops.csv`. Then start at Question 2.1.

```
set.seed(2102)
n <- 400
Location  <- sample(c("Downtown", "Suburb", "Campus"), n, replace = TRUE,
                    prob = c(0.35, 0.40, 0.25))
DriveThru <- sample(c("Yes", "No"), n, replace = TRUE, prob = c(0.4, 0.6))
Chain     <- sample(c("Yes", "No"), n, replace = TRUE, prob = c(0.45, 0.55))
Staff     <- sample(3:15, n, replace = TRUE)
Price     <- round(4.5 + 0.6 * (Location == "Downtown") - 0.4 * (Chain == "Yes") +
                   rnorm(n, 0, 0.5), 2)
Age       <- round(ifelse(Location == "Campus", rnorm(n, 22, 2.5), rnorm(n, 38, 7)))
Revenue   <- round(20 + 2.5 * Staff - 3 * Price + 6 * (Location == "Downtown") +
                   5 * (DriveThru == "Yes") + rnorm(n, 0, 5), 1)
score     <- Revenue + rnorm(n, 0, 6)
Rating    <- ifelse(score < quantile(score, 0.3), "Poor",
                    ifelse(score < quantile(score, 0.8), "Average", "Excellent"))
coffee <- data.frame(Shop_ID = 1:n, Revenue, Price, Staff, Age,
                     Location, DriveThru, Chain, Rating)
write.csv(coffee, "coffee_shops.csv", row.names = FALSE)
```